

Local Anesthetic Systemic Toxicity

Section 1: Case Summary

Scenario Title:	Local Anesthetic Systemic Toxicity Case
Keywords:	local anesthetic toxicity, lipid emulsion therapy
Brief Description of Case:	A femoral nerve block has just been performed on a 65-year-old male who sustained a right femoral neck fracture. The patient also sustained a laceration to the scalp which was repaired by the medical student. The patient is now complaining of blurry vision, paresthesias, and "twitchiness". The patient progresses on to seizure followed by cardiac arrest and will need high quality ACLS care along with lipid emulsion therapy.

Goals and Objectives	
Educational Goal:	To allow learners to identify and treat a severe overdose in a patient who received local anesthetic for a peripheral nerve block.
Objectives: (Medical and CRM)	Medical 1. Identify local anesthetic toxicity 2. Provide high quality ACLS care 3. Initiate lipid emulsion therapy for treatment of local anesthetic toxicity CRM 1. Manage resources to complete simultaneous investigation and treatment of an unstable patient including poison control and relevant consultants 2. Delegate tasks effectively to facilitate resuscitation
EPAs Assessed:	ME 2.4 Initiate medical treatment of the patient with an overdose, toxic ingestion or exposure, including specific antidote therapy

Learners, Setting and Personnel						
Target Learners:	☒ Junior Learners		☒ Senior Learners		☐ Staff	
	☒ Physicians		☒ Nurses		☐ RTs	☐ Inter-professional
	☐ Other Learners:					
Location:	☒ Sim Lab		☐ In Situ		☐ Other:	
Recommended Number of Facilitators:	Instructors: 1					
	Confederates: 0-1					
	Sim Techs:					

Scenario Development	
Date of Development:	13/10/2019
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: George Geoparodopolous		Age: 65		Gender: Male	Weight: 80 kg
Presenting complaint: Palpitations and funny taste in mouth					
Temp: 37.0°C	HR: 78/min	BP: 156/78	RR: 22/min	O ₂ Sat: 99%	FiO ₂ :
Cap glucose: 7.0 mmol/L			GCS: 15 (E4 V5 M6)		
Progress note (to be used as stem to be read to learner): 65-year-old male presents after sustaining a fall in the shower. He has a femoral neck fracture and large laceration on his scalp. A medical student has repaired his laceration. You are called to the bedside as the patient has started to complain of blurry vision, paresthesias and "twitchiness".					
Allergies: NKA					
Past Medical History: A-fib Diabetes mellitus type II Chronic kidney disease			Current Medications: Warfarin Metformin Ramipril		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, confederate, SP, etc.)?</i>	
Further HPI to be provided if the resident asks: The patient just received a femoral nerve block by the staff MD 10 minutes prior.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: S1, S2 but sounds irregular at times	Neuro: Alert, complaining of blurry vision, paresthesias, and "twitchiness"
Resp: GAEB, no adv sounds	Head & Neck: Scalp laceration repaired
Abdo: soft, non-tender	MSK/skin: Cannot move left leg, but neurovascularly intact
Other: No distress but expressing anxiety about sudden change in vision	



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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Gloves Stethoscope Defibrillator NIBP cuff / Pulse oximeter IV Bags/Lines IV Push Medications Nasal Prongs Non-Rebreather Mask Bag Valve Mask Laryngoscope ET Tubes
C. Required Medications
ACLS meds Lipid emulsion 20%
D. Moulage
Bandage on the head. Right leg looks slightly shorter and externally rotated than the left leg.
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Rhythm: Normal sinus rhythm with some intermittent 1 st degree blocks

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Nurse	Tells you that patient complaining of palpitations, paresthesias, and “twitchiness”

Simulation Scenario Template

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	Facilitator Notes
1. Baseline State Rhythm: Prolonged PR on monitor HR: 78/min BP: 156/78 RR: 22/min O ₂ SAT: 99% T: 37°C	Alert, complaining of blurry vision, paresthesias, and twitchiness.	<u>Expected Learner Actions</u> ☐ History and physical ☐ Monitors ☐ Ask for med list ☐ 2 nd peripheral IV ☐ Get EKG ☐ Ask for medications and doses given up until now and checks glucose ☐ Verbalize differential diagnosis including local anesthetic toxicity and head injury	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - If asks for lipid emulsion, will be told it needs to be prepared - If calls poison control, will be prompted to talk to them over the phone by nurse who will get back to them - If asks for what medications given, lidocaine by med student but not sure how much as well as bupivacaine 45mL of 0.5% <u>Triggers</u> <i>For progression to next state</i> - All actions complete or 5 min into case → 2. Progressive CNS toxicity
2. Progressive CNS toxicity Rhythm: Difficult to tell due to seizure activity HR: 120/min BP: 180/90 RR: 40 /min O ₂ SAT: 85 % T: 37.0°C	Patient starts seizing	<u>Expected Learner Actions</u> ☐ Recognizes seizure ☐ Lorazepam 0.1mg/kg)max 4mg) IV or midazolam 10mg IV (repeat q5mins) ☐ Check glucose, send lab panel, consider broad differential diagnoses ☐ Puts on high flow oxygen and considers intubation ☐ Lipid emulsion 20% Bolus: 1 - 1.5 ml/kg over 1 min (Can	<u>Modifiers</u> - Benzodiazepines given → Seizure continues - Supplemental O ₂ → O ₂ SAT: 95 % - At 9 minutes → Poison control calls back and says if seizing or cardiac arrest, give lipid emulsion 20% bolus 1-1.5 mg/kg over 1 min then infusion <u>Triggers</u> - All actions complete or 10 minutes → 3. Cardiac arrest



Simulation Scenario Template

		repeat bolus every 3 minutes up to a total dose of 3 ml/kg) THEN ☐ 0.25 mL/kg/minute, and repeat bolus as necessary		
3. Cardiac Arrest HR: VT at 220/min BP: 0 RR: 0/min O ₂ SAT: 70% No pulse	Patient unresponsive	<u>Expected Learner Actions</u> ☐ Recognizes cardiac arrest ☐ High quality ACLS care ☐ Consider bicarbonate ☐ Consider intubation ☐ Give lipid emulsion at this point if not done already ☐ Call ICU +/- consideration for ECMO	<u>Modifiers</u> <u>Triggers</u> - If does not give lipid emulsion continue ACLS until → End scenario - After lipid emulsion → 4. ROSC	
4. ROSC HR: 90/min BP: 100/70 RR: 16 /min O ₂ SAT: 95% Strong pulse	Patient moaning	<u>Expected Learner Actions</u> ☐ Identify ROSC ☐ Get EKG + full head to toe reassessment and vital signs ☐ Give bolus 500 cc NS IV for hypotension	<u>Triggers</u> - Calls ICU for transfer or after 15 minutes total → End Scenario	

Simulation Scenario Template

Appendix A: Laboratory Results

Only VBG is back in time (drawn prior to seizure pH 7.4, pCO₂ 40, pO₂ 98, HCO₃ 24, lactate 2.6



Simulation Scenario Template

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



Source: <https://radiopaedia.org/cases/femoral-neck-fracture-1>

Simulation Scenario Template

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Debriefing as a group, without video.

Sample questions for debriefing:

CRM

1. How can you use the team to effectively communicate with resources (e.g. poison control) while managing an unstable patient?
2. How do you ensure you have gathered all the information you want while managing an unstable patient?

Medical Expert

1. What are the initial symptoms of LAST? Does it differ with lidocaine vs. bupivacaine? What is the toxic dose of each of these?
2. What considerations in managing seizures secondary to LAST?
3. What changes in ACLS medications should be considered when managing LAST?

Key moments:

1. Recognizing LAST.
2. Calling poison control early.
3. Giving lipid emulsion which is the definitive treatment for LAST while also providing supportive management.

References

1. LAST – REBEL EM – Emergency Medicine Blog retrieved from: <https://rebelem.com/local-anesthetic-systemic-toxicity-last/>
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